

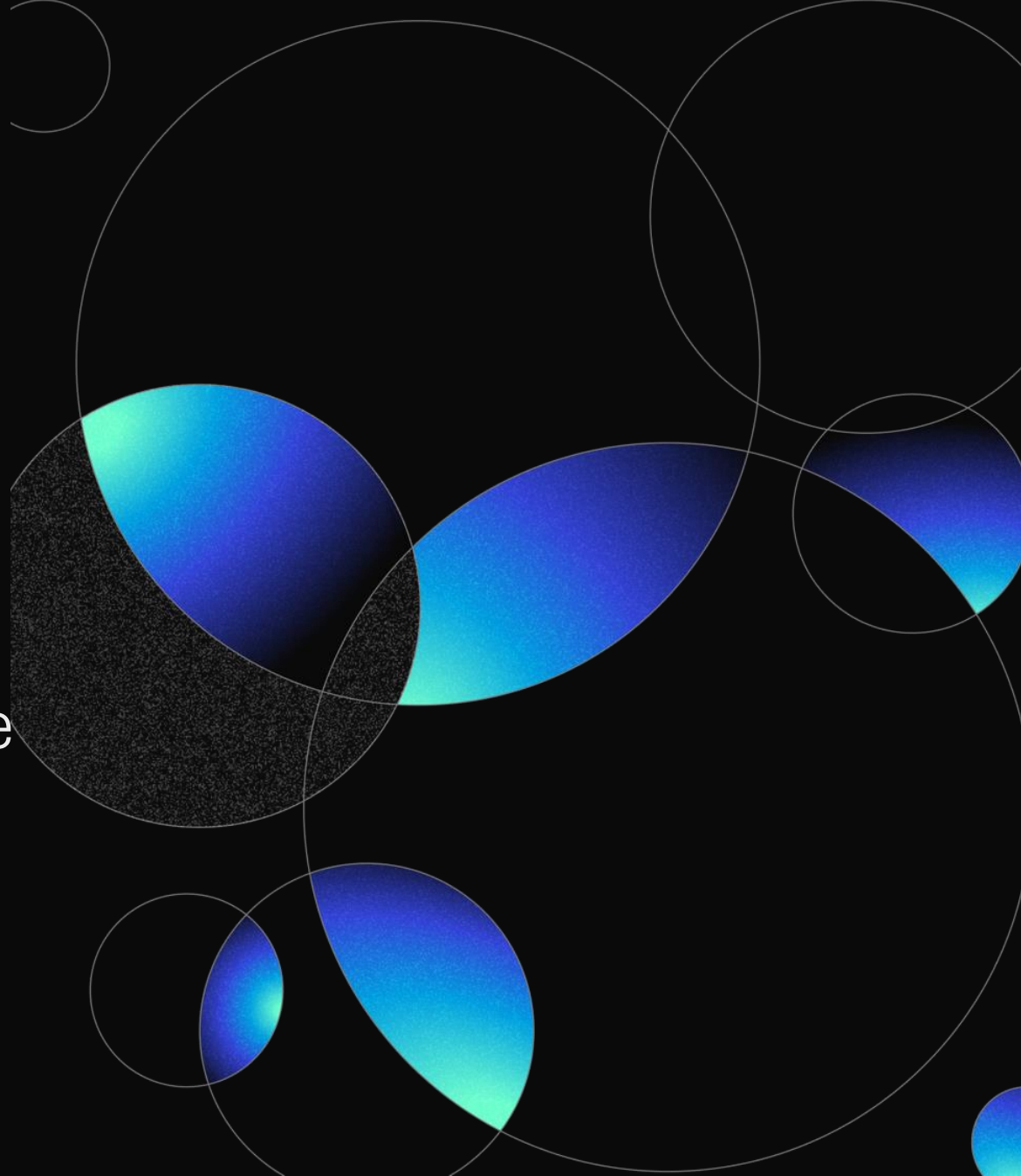


Trusted AI for Defense Decisions

Mathematical Optimization in Practice



Chung-Kyun Han
Technical Account Management - APAC



Agenda



- 01 Mathematical Optimization 101
- 02 Example Applications
in Military & Defense
- 03 Optimization as “Trusted AI”
- 04 Our History
- 05 The Road Ahead

Mathematical Optimization 101

Museum Heist Example



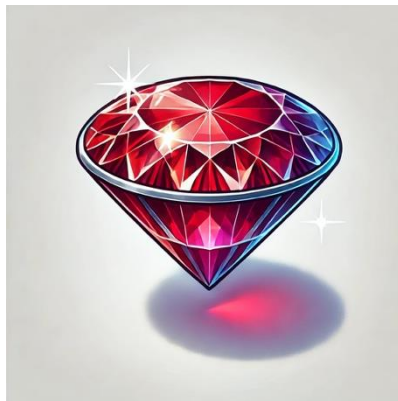
\$13m
8kg



\$11m
6kg



\$7m
3kg



\$2.5m
2kg



\$10m
5kg



Capacity = 11kg

Museum Heist Example



\$13m
8kg



\$11m
6kg



\$11m
6kg



\$2.5m
2kg



\$2.5m
2kg



\$2.5m
2kg



\$10m
5kg



\$7m
3kg



11kg



8kg

What is Mathematical Optimization?



\$13m
8kg

x_1



\$11m
6kg

x_2



\$2.5m
2kg

x_3



\$10m
5kg

x_4



\$7m
3kg

x_5



11kg

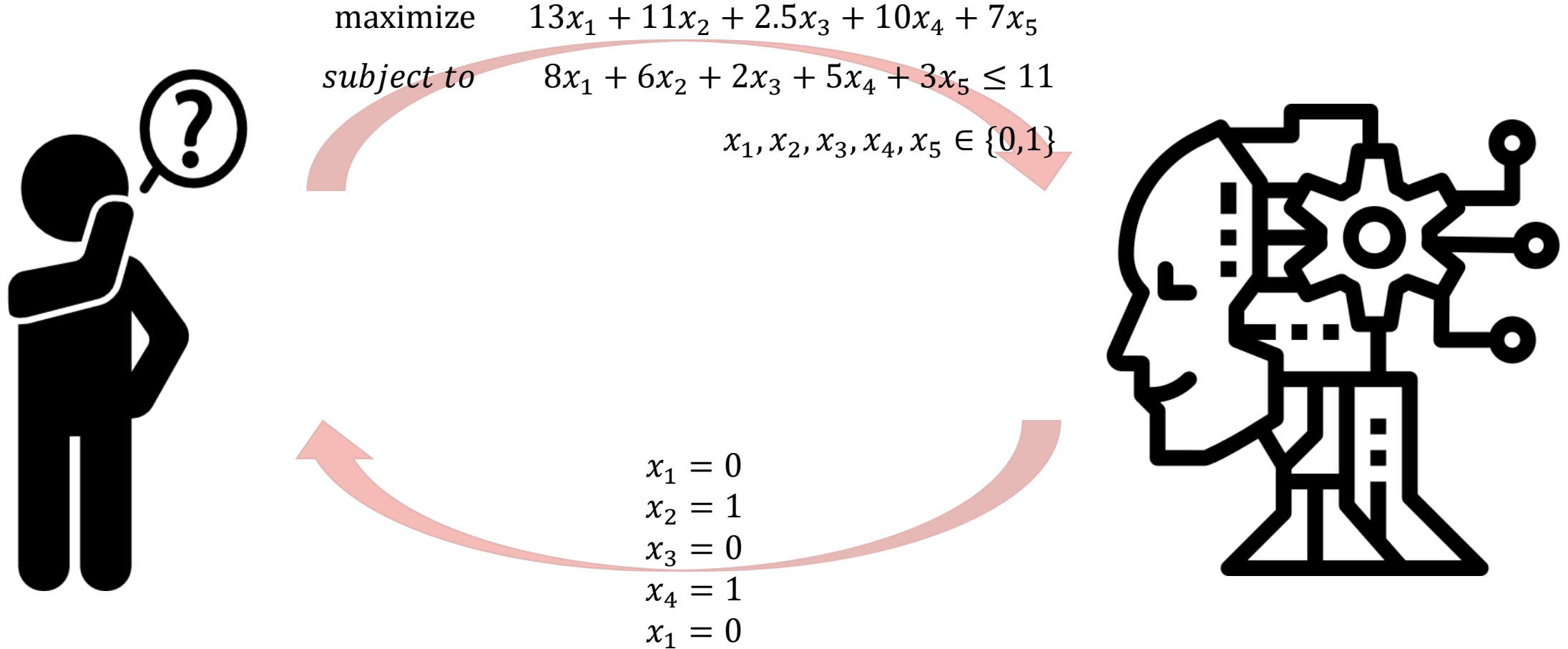
Maximize the value of the items taken

Weight of items in bag respect capacity

Should I include this item in my bag?

$$\begin{aligned} &\text{maximize} && 13x_1 + 11x_2 + 2.5x_3 + 10x_4 + 7x_5 \\ &\text{subject to} && 8x_1 + 6x_2 + 2x_3 + 5x_4 + 3x_5 \leq 11 \\ &&& x_1, x_2, x_3, x_4, x_5 \in \{0,1\} \end{aligned}$$

Mathematics as a language



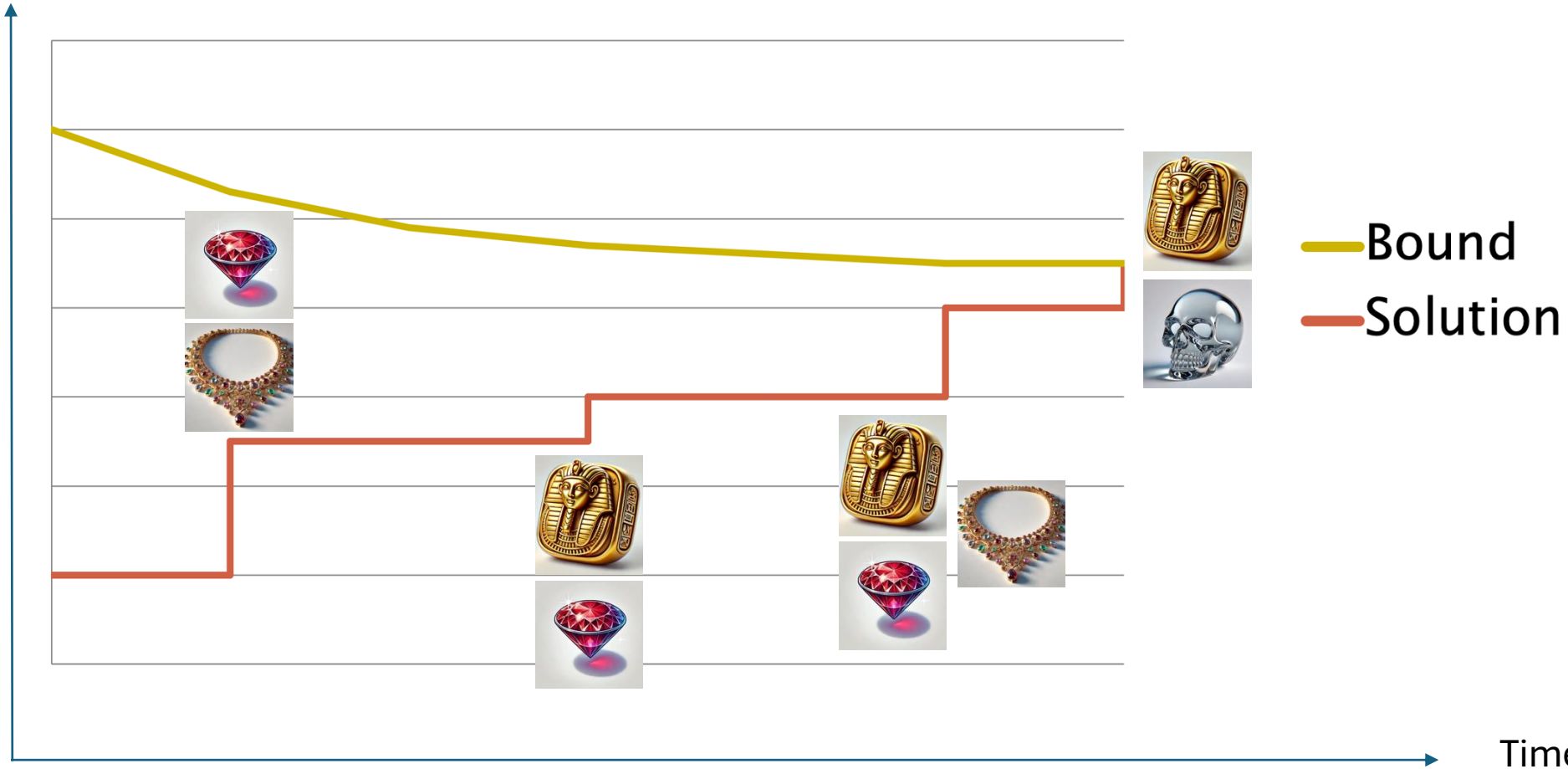
Turning models into code is the easy part

$$\begin{aligned} &\text{maximize} && 13x_1 + 11x_2 + 2.5x_3 + 10x_4 + 7x_5 \\ &\text{subject to} && 8x_1 + 6x_2 + 2x_3 + 5x_4 + 3x_5 \leq 11 \\ &&& x_1, x_2, x_3, x_4, x_5 \in \{0,1\} \end{aligned}$$

```
1  import gurobipy as gp
2  from gurobipy import GRB
3
4  m = gp.Model()
5  painting = m.addVar(vtype=GRB.BINARY)
6  statue = m.addVar(vtype=GRB.BINARY)
7  gem = m.addVar(vtype=GRB.BINARY)
8  skull = m.addVar(vtype=GRB.BINARY)
9  necklace = m.addVar(vtype=GRB.BINARY)
10
11 m.ModelSense = GRB.MAXIMIZE
12 m.setObjective(13*painting + 11*statue + 2.5*gem + 10*skull + 7*necklace)
13 m.addConstr(8*painting + 6*statue + 2*gem + 5*skull + 3*necklace <= 11)
14 m.optimize()
15 print(f"Optimal objective is {m.ObjVal}")
16
```

Solutions can be proven optimal

Objective Value

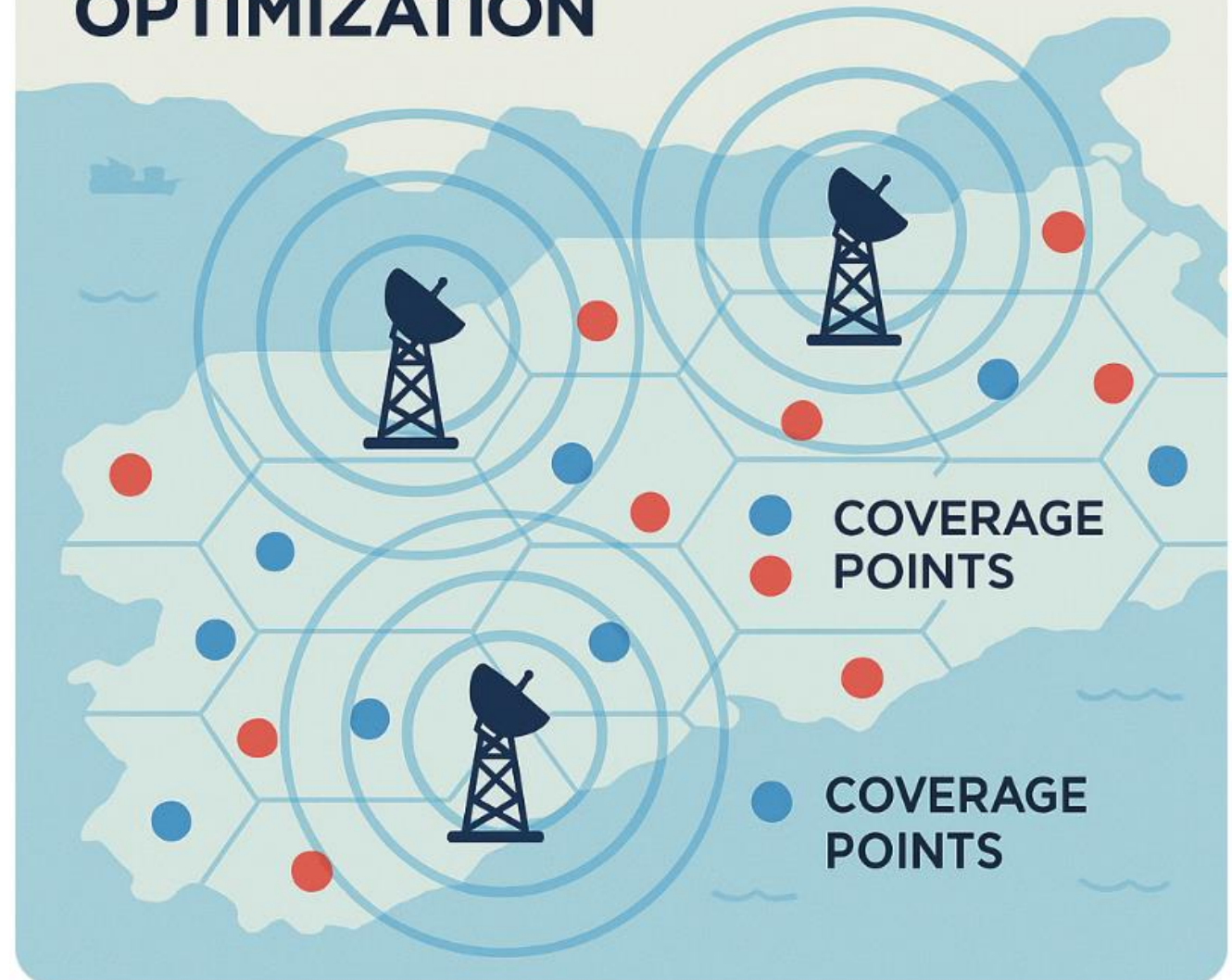


Example Applications in Military & Defense

Channel & Radar Allocation

- Allocate limited communication channels and radar resources across diverse maritime zones and operational scenarios
- Optimization problem types
 - Set Covering Problem
 - Maximum Coverage Problem
- Mathematical model: optimal resource allocation
 - Each radar can cover points within a specific radius
 - Candidate locations for deployment are given
 - Cover all surveillance target points, or maximize coverage with a limited number of radars

COMMUNICATIONS/RADAR RESOURCE ALLOCATION OPTIMIZATION



| Training & Education Scheduling

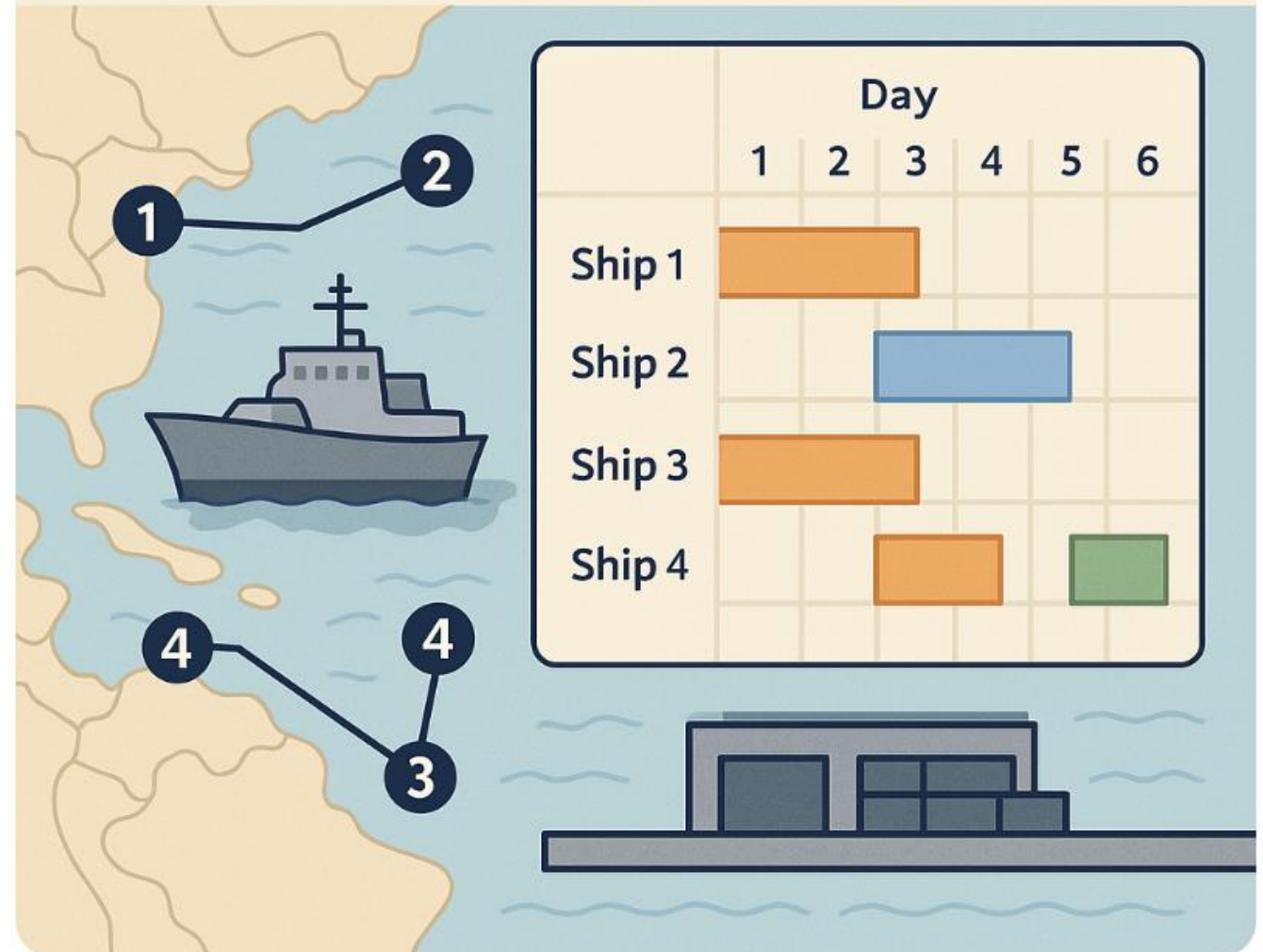
- Design training schedules considering diverse training content, locations, personnel, and vessel availability
- Optimization problem types
 - Multi-objective scheduling
 - Resource Constrained Project Scheduling Problem
- Mathematical model: optimal allocation of personnel, vessels, and locations
 - Each training course must run continuously for a set duration
 - Each course requires different resources (instructors, locations, vessels, etc.)
 - Resources cannot be assigned to multiple trainings at the same time
 - The goal is to conduct as many trainings as possible, or maximize the total priority score



Maintenance & Docking Scheduling

- Optimize maintenance and docking schedules under resource constraints while maximizing vessel availability
- Optimization problem types
 - Preventive Maintenance Scheduling
- Mathematical model: optimal maintenance plan under resource constraints
 - Maintenance must be performed on multiple vessels over a given period
 - Each vessel must be serviced within a set window, with limited maintenance time and docks
 - Complete all maintenance by deadline while avoiding docking-resource conflicts and minimizing availability loss

MAINTENANCE AND DOCK SCHEDULING OPTIMIZATION



- # ARTILLERY FIRE SCHEDULE OPTIMIZATION
-
- The diagram illustrates the optimization of artillery fire schedules for four units (1, 2, 3, 4) across six time steps (1 to 6). The units are positioned in a mountainous landscape, and the targets are marked with red bullseyes.
- Legend:**
- 1: Prior #1
 - 2: Sequencing
 - 3: Prior #2
 - 4: Prior #3
- Fire Schedule Optimization Table:**
- | Unit | Time Step | Task / Event |
|--------|-----------|-------------------|
| Unit 1 | 1 | T1 (Prio 1) |
| Unit 1 | 2 | Continuous Firing |
| Unit 2 | 2 | T1 (Prio 2) |
| Unit 2 | 3 | Continuous Firing |
| Unit 3 | 3 | T2 (Prio 2) |
| Unit 3 | 4 | Continuous Firing |
| Unit 4 | 4 | T3 |
| Unit 4 | 5 | Continuous Firing |
| Unit 4 | 6 | T4 |
- Total Completion Time Minimization**
- The diagram also shows the synergistic simultaneous fire effect, where multiple units fire on the same target simultaneously, resulting in a larger fire effect (indicated by a larger flame icon).

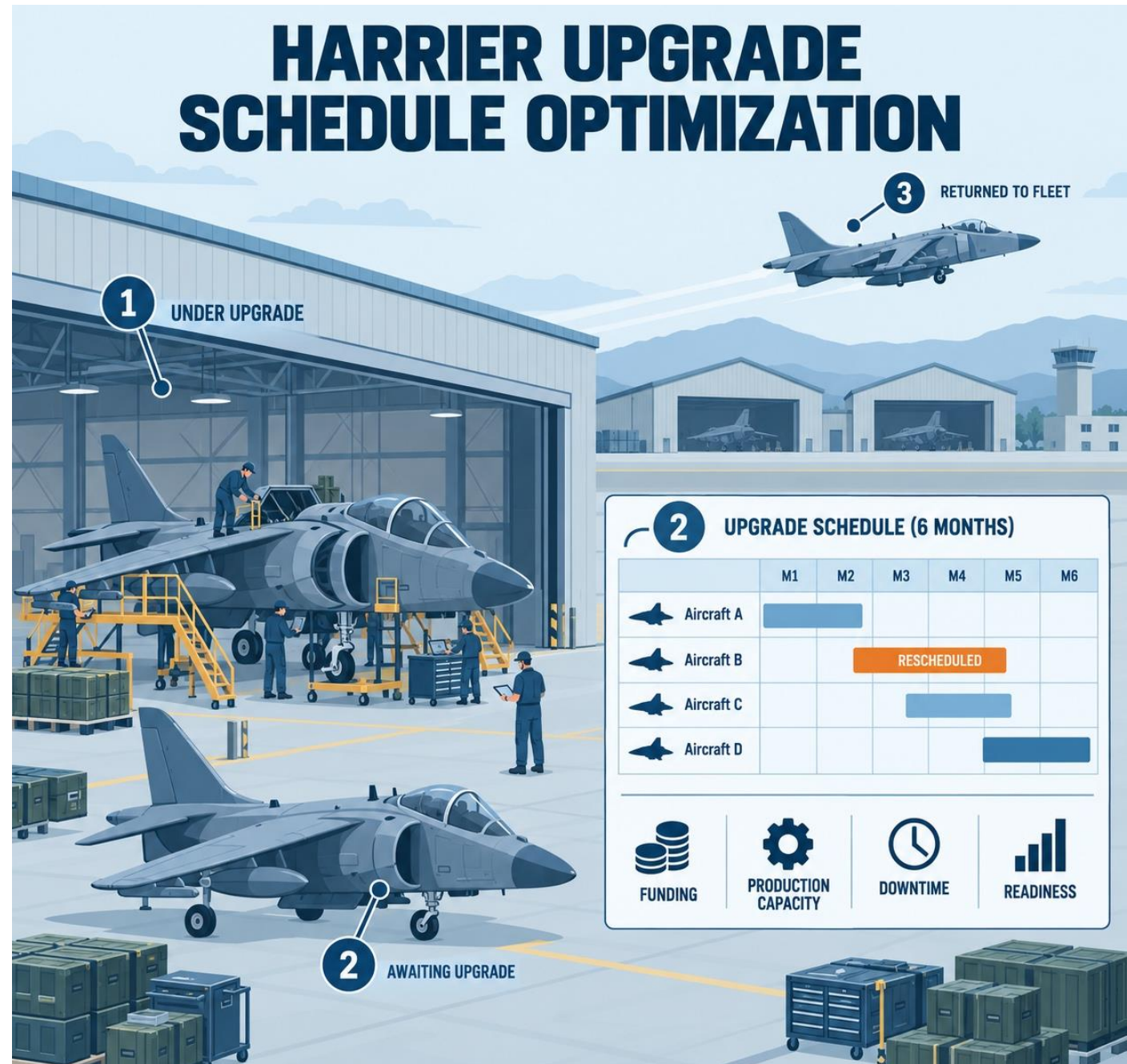
Logistics Packing Optimization

- Pack pallets and transports at a Deployed Military Logistics Hub (Dstl with Decision Lab, LOGOS)
- Optimization problem types
 - Bin packing / knapsack for pallet packing
 - Transport loading with multi-drop routing
- Mathematical model: maximum load per pallet and transport
 - Item size, weight, fragility, and required orientation
 - Destination and drop order on multi-drop loads
- Business impact
 - About 20% more items packed, and faster, in field trials
 - Up to 20% fewer transports at larger transport scale
 - Solutions in tens of seconds to minutes



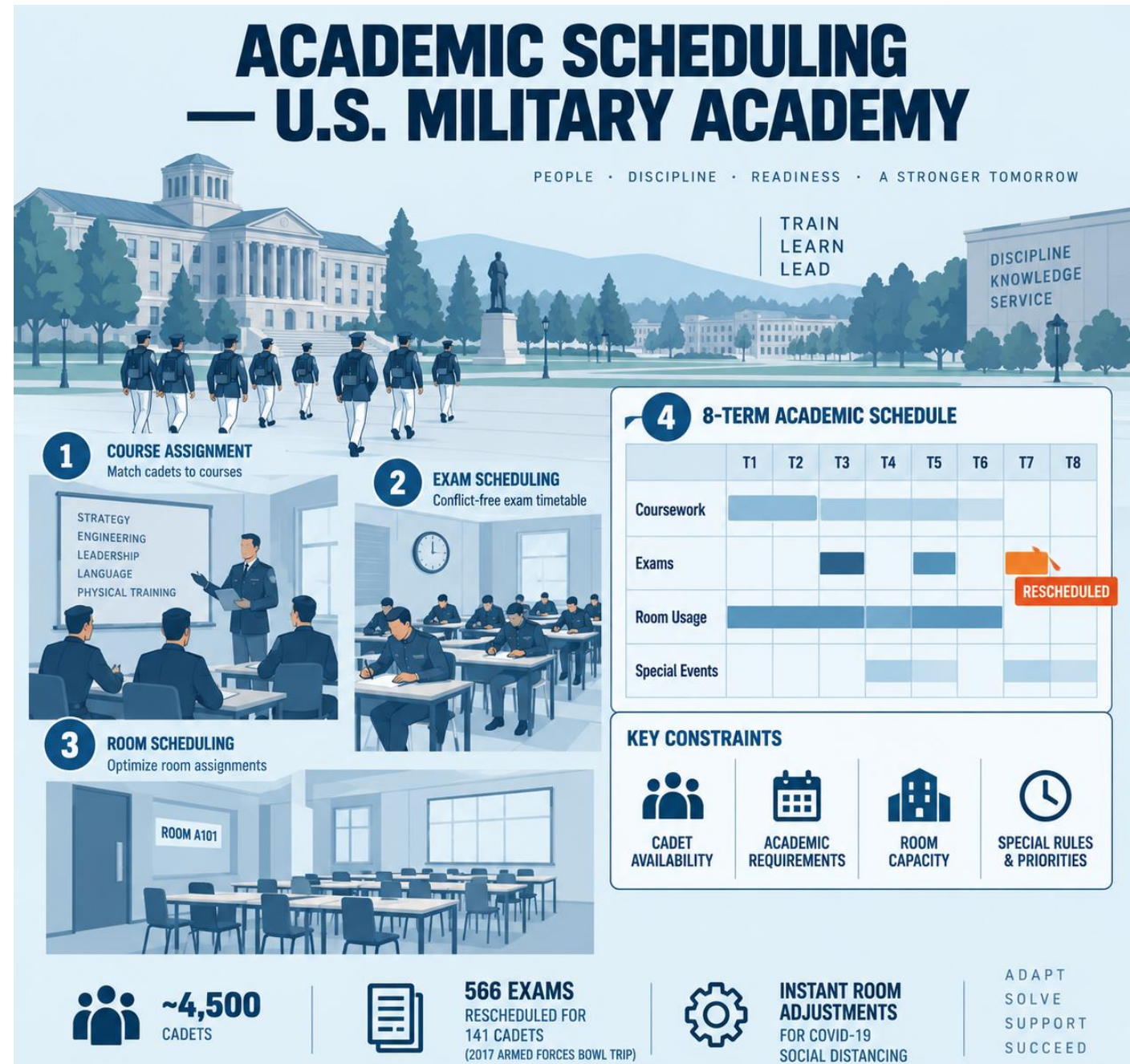
Aircraft Upgrade Scheduling

- Schedule AV-8B Harrier upgrades across sites and organizations (US Marine Corps with CNA)
- Optimization problem types
 - Resource-constrained project scheduling
- Mathematical model: keep the most aircraft mission-capable
 - Funding and production capacity limits
 - Downtime caps how many jets are out at once
 - Reschedules on delays and design changes
- Business impact
 - Peak mission-capable fleet throughout
 - Fewer missed milestones
 - Fewer contract extensions and modifications



Academic Scheduling — U.S. Military Academy

- Optimize the 8-term academic schedule for ~4,500 cadets across coursework, exams, and room assignments
- Optimization problem types
 - Assignment + Scheduling
- Mathematical model: multiple competing objectives optimized under numerous prioritizable business rules
 - Regular term scheduling
 - Term-end exam scheduling
 - Room scheduling
- Business impact
 - Rescheduled 566 exams for 141 cadets on short notice (2017 Armed Forces Bowl trip)
 - Instantly adjusted room assignments for COVID-19 social distancing



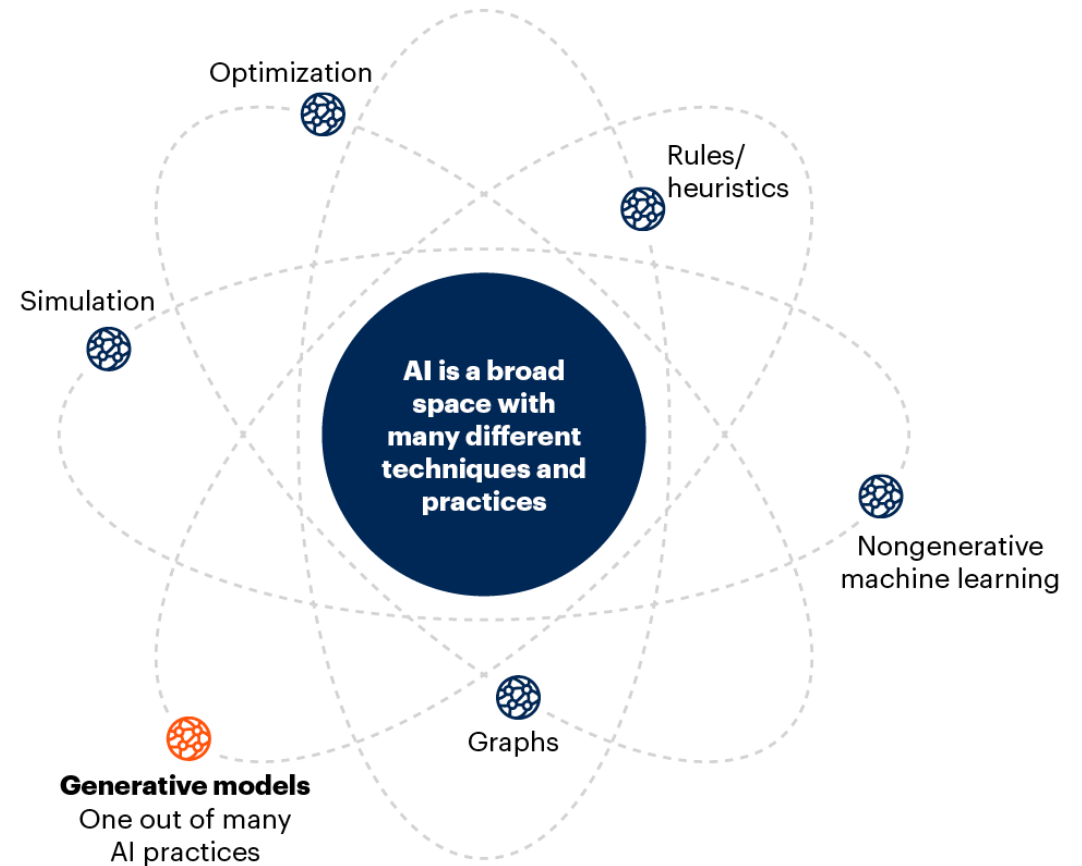
Optimization as Trusted AI

Defining AI

Per **Gartner**...

“Artificial intelligence (AI) applies advanced analysis and logic-based techniques to interpret events, and support and automate decisions and take actions.”

AI Does Not Revolve Around GenAI



Source: Gartner
806238_C

Gartner IT Glossary, Artificial Intelligence (AI), 16 October, 2024,
<https://www.gartner.com/en/information-technology/glossary/artificial-intelligence>

„Intelligence“

Human Intelligence



Learning

Logic

Problem-Solving

Creativity

Deciding

Artificial Intelligence



Artificial Intelligence is the capacity of computer systems to exhibit or simulate intelligent behavior by processing data and performing tasks typically associated with human intelligence.

AI for Decision Intelligence

Decision Intelligence is the tailored combination of the right data, the right AI disciplines and the right human influence in a business decision-making process.

- **Learning** using **Machine Learning**
Generation of insights from business data
- **Logic** using **Algorithmic Reasoning**
Generation of conclusions (logical inference) through algorithms
- **Creativity** using **Generative AI**
Generation of new content from a natural language prompt based on learned structure
- **Problem-Solving** using **Forecasting and Simulation**
Generation of predictions and scenarios
- **Deciding** using **Mathematical Optimization**
Generation of optimal planning outcomes

Why Can You Trust Optimization?

Optimization

Mathematically **correct** and **provably** optimal



Deterministic and **reproducible**



Constraints are **explicit** and **inviolable**, states when no feasible solution exist



Quantifies **trade-off** and allows **sensitivity** analysis



ML / Gen AI



Outputs are **plausible**, not provably correct



Sensitive to training data **drift** or prompt **phrasing** (LLMs)



Constraints are learned/inferred, can be ignored and violated, or **hallucinate** compliance



Confidence $\neq$ Correctness

Optimization is AI that...



Is built for decision making

The building blocks and structure of optimization are the same as many decisions

Compliments machine learning and generative models

Not all problems are a nail, so you need more than a hammer

Provides explainability, trust and value

The declarative and deterministic nature of optimization allows for justifiable and auditable outputs

Is a key component of Decision Intelligence

Rigorous mathematical framework that gives confidence to business decisions

Trusted Research & Development

Our History



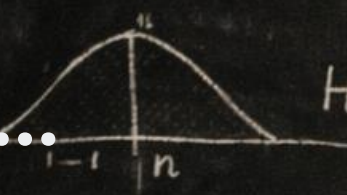
Back to 1939...

$$\sigma^2 = \sum_{n=1}^n \frac{(X - \bar{X})^2}{n}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$E = \sum_{n=1}^n x_n$$

$$\frac{1}{2} =$$



$$H_0: \mu_1 = \mu_2$$

$$E(x) = \sum_{n=1}^n x P(x)$$

$$\sum_{F=1}^n = \sum_{F=3}^n \hat{P}(x)$$



$$P(x \leq a) = \int_a^{\infty} f(x) dx$$

$$= \int_c^{\infty} \frac{\sqrt{X} - \mu^2}{n}$$

$$\sigma^2 = \sum_{n=1}^n \frac{(X_i - \bar{X})^2}{n}$$

$$\sigma = \sqrt{\sum_{n=1}^n \frac{(X - \mu)^2}{n}}$$

$$t = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

$$\sum_{n=0}^{\infty} = \sum_{n=0}^{\infty} (X_i - \mu)$$



$$\bar{\sigma} = \sum_{n=5}^n \frac{(\bar{X} - \bar{\mu})^2}{n}$$

$$P_B = \frac{1}{2} \leq -\frac{1}{2} R$$

1939

George Dantzig solved two problems he thought were homework, only to discover they were long-unsolved statistical challenges.

His story shows that what we believe is possible can shape what we achieve.



1948

“A certain wide class of practical problems appears to be just **beyond the range of modern computing machinery**. [...]

Typically, these problems involve a complex of different activities in which one wishes to know which activities to emphasize in order to carry out desired objectives under known limitations.”

“Programming in a linear structure,”

George B. Dantzig, published August 1948 by U.S. Air Force Comptroller



1948 - 1988 – 4 decades of R&D



1952-54 LP code on "Card Programmable Calculator"

1971 Supersparsity, P4, Pre-assigned pivot factorization

1979 Khachian's paper on the "ellipsoid method"

1984 Karmarkar's paper on interior-point methods

1972-1985: Very theoretical research in OR by Bixby et al ("matroid theory")

The 90's

1988 - Robert E. Bixby founded CPLEX Optimization

A lot of development in the 1990s

- LP performance took off (Primal-dual log-barrier algorithms completely reset the bar)
- Data became plentiful and accessible
- ERP systems became commonplace
- Popular new applications begin to show that integer programming could work on difficult, real-world problems (airlines, supply chain, etc)

TEST RUNS*

NETLIB PROBLEM	# Rows	# Cols	# #0	LOPT		XMP	
				# ITER	Time	# ITER	Time
GFRD-PNC	616	1092	2378	628	80.5	983	144.1
SCRS8	490	1169	3183	735	112.2	1271	182.7
SIERRA	1227	2036	7303	493	93.8	950	270.0
STANDAT	369	1183	3032	181	14.7	75	8.1
SCAGR25	471	500	1555	517	69.1	1470	248.3
SHARE2B	97	79	695	93	4.7	138	6.5
SHARE1B	118	225	1152	217	16.1	411	28.8
E226	224	282	2579	444	50.8	655	70.2
CAPRI	272	353	1768	364	32.4	550	14.5
BANDM	306	472	2495	362	54.5	1679	250.9
STAIR	357	463	3847	500	76.1	1117	530.9
ETAMACRO	401						144.
SHIP2L	1152						515.7

* Run on SUN 3/ -O end -f68



Progress in LP (1988-2004)

Operations Research, Jan 2002, pp. 3—15, updated in 2004



Algorithms (machine independent): **3,300x**

Primal versus best of Primal/Dual/Barrier

Machines (workstations → PCs): **1,600x**

NET: Algorithm × Machine **5,300,000x**

(2 months/5300000 ≈ 1 second)

2009 - Public Gurobi Launch at INFORMS

Gurobi Optimization

- ▶ Gurobi Optimization, Inc. founded July, 2008
- ▶ Building a suite of products:
 - ▶ LP (simplex) and MILP in version 1.0
 - ▶ Barrier, QP, QCP, MIQP, and MIQCP later
- ▶ Gurobi available through partners January
- ▶ Gurobi standalone available 6 May 2009



9-Mar-26

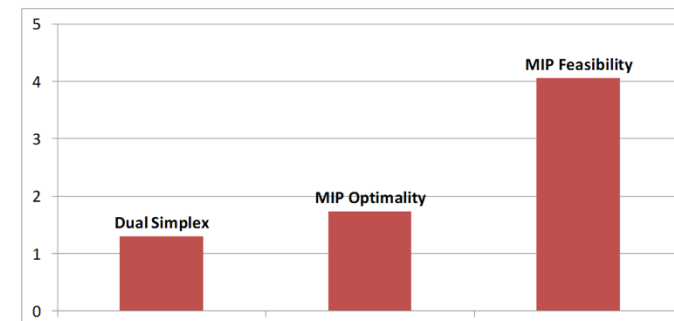
© 2009 Gurobi Optimization

3

Building From the Ground Up

- ▶ Start from a clean slate
 - With the benefit of 20+ years of experience
- ▶ Things have changed:
 - Two examples:
 - “Sub-MIP” as a pervasive approach
 - Ubiquitous parallel processing

Performance Summary – Gurobi 2.0 vs CPLEX 12.1



9-Mar-26



9-Mar-26

© 2009 Gurobi Optimization

11
4



10 Years Ago (2016) - Our Team

The Hardware & Software Performance Race

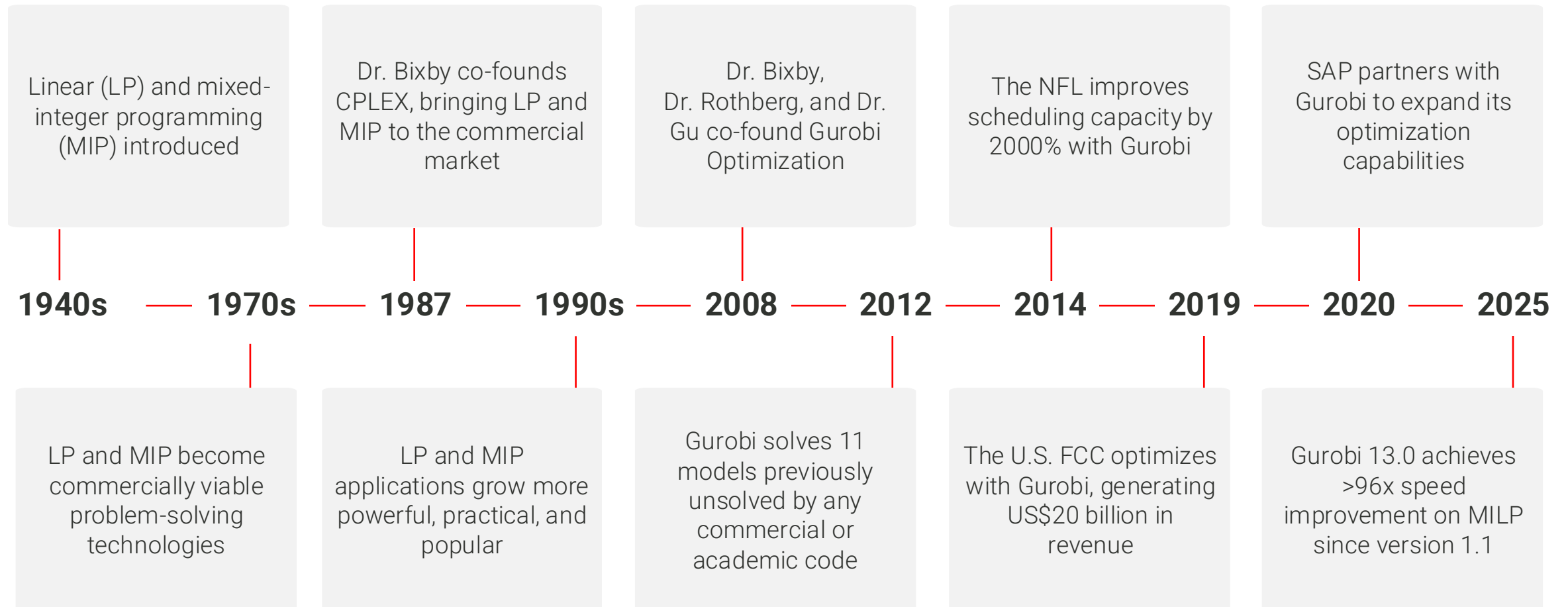


- **Single threaded CPU performance**
1991-2022: **$\approx 1,542x$ speed-up**
- **MIP Algorithm/Software performance**
1991-2022: **$\approx 2,035,000x$ speed-up**
- **Performance gain CPU & MIP Optimization**
1991-2022: **$\approx 3,138,000,000x$ (3.14bn) speed-up**

Mathematical illustration: An Optimization business problem which would have taken **100 years in 1991** to solve would do so **today in 1s**.

Sources: R. E. Bixby (Gurobi Optimization), "Computational Progress in Linear and Mixed Integer Programming"; Solver Max, "Solver performance: 1989 vs 2024."

70+ years of problem-solving innovation







UNTIL NOW.

Gurobi is the most powerful - fastest, versatile, and reliable - mathematical optimization solver which enables confident decision-making at scale.



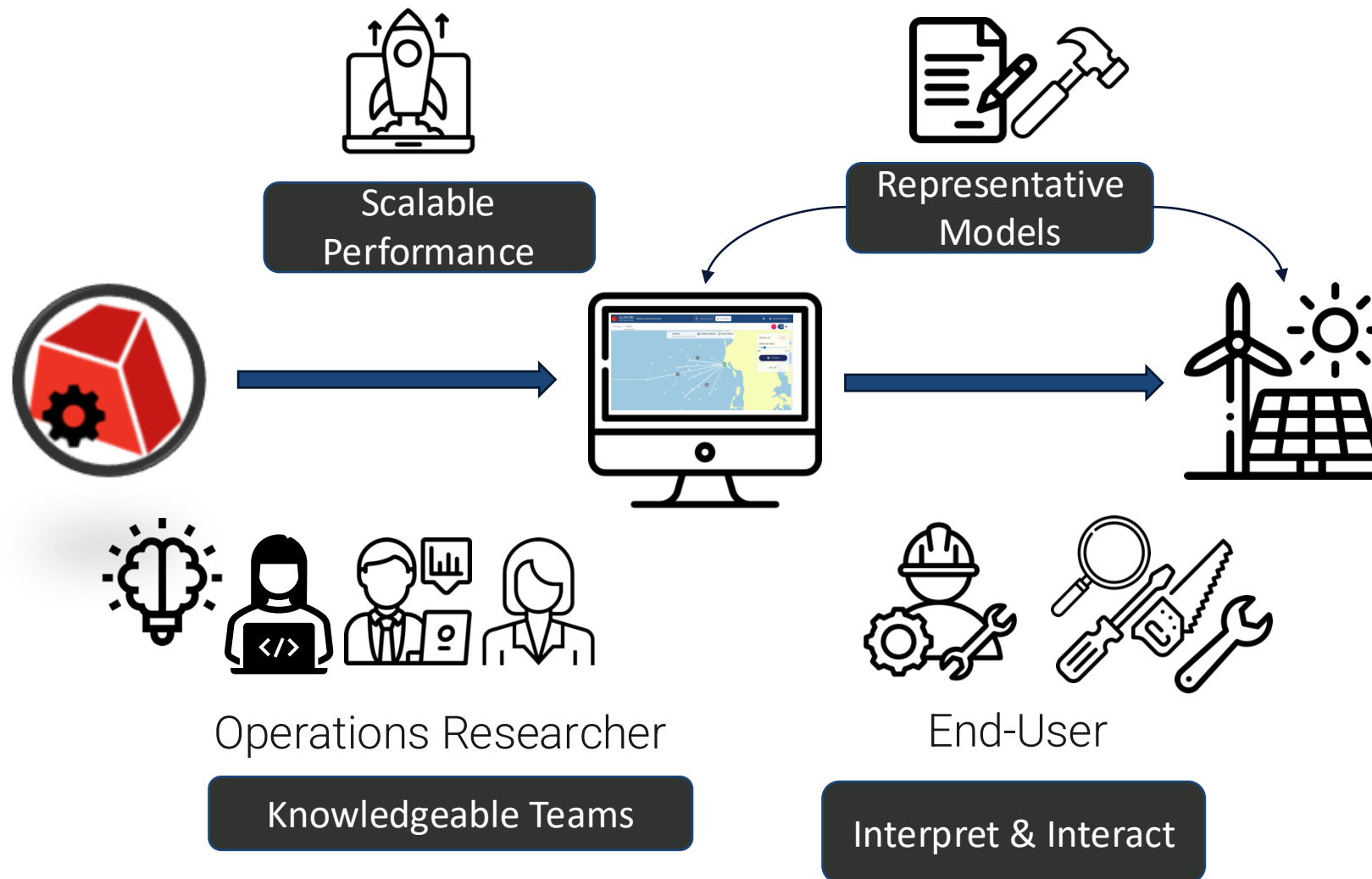
LOOKING FORWARD.

Gurobi powers high-impact decisions with trusted AI, setting the standard in decision intelligence through unmatched performance, productivity, and deployment capabilities.

The Road Ahead

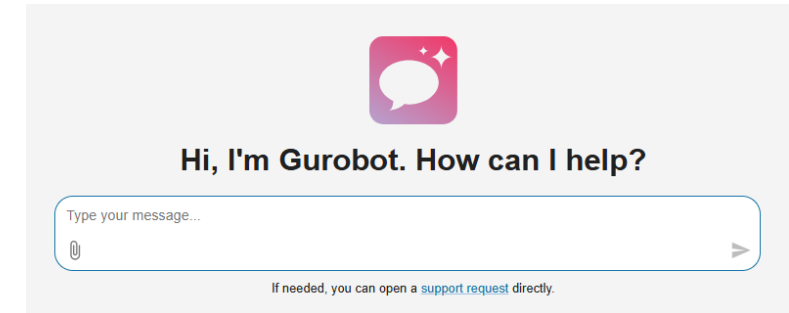
Our Intelligence Vision

GenAI will make Optimization more accessible



Gurobot – Your Always-On Gurobi Expert Agent

Gurobot is our support powerhouse designed to streamline your experience and put expert guidance right at your fingertips, with:



- **Instant Answers and Assistance:** Real-time help with troubleshooting, licensing, and more.
- **Smart Resource Links:** Gurobot connects you to the most relevant documentation
- **Built-in Best Practices:** Responses are backed by Gurobi's knowledge base
- **Code Snippets on Demand:** Tailored, ready-to-run gurobipy snippets
- **Enterprise-Grade Security:** Powered by Anthropic's Claude Foundation LLM and securely hosted on AWS, Gurobot is designed to safeguard your data.

Explainer Preview



Analysis Complete

I've analyzed the infeasibility in your dairy processing model and created a comprehensive report. Here's a summary of the key findings:

Core Problem

Your model is infeasible because it requires producing at least **20,000 units of Premium_drink**, but the system's constraints prevent generating sufficient quantities of the necessary ingredients. The main conflicts are:

1. **Capacity Bottlenecks:** The protein concentration capacity is limited to 10,000 units, which can produce at most ~6,000 units of protein concentrate (with the best yield of 0.6)

Type your message...



The Gurobi AI Agent is using generative AI and may produce inaccuracies. By using this service, you agree to our [Privacy Policy](#) and [Terms of Service](#).

You have used 4% of your daily limit.

infeasibility_analysis_report.md (10.3 KB) ▾

COPY

DOWNLOAD



Infeasibility Analysis Report

File Analyzed: moodel_infeasible.ilp

Analysis Date: 2026-04-08

Generated For: Caroline Weinberg

1. IIS Introduction

This model represents a dairy processing optimization problem with multiple production stages: milk separation, protein concentration, secondary processing, and final product blending. The IIS (Irreducible Inconsistent Subsystem) contains the following key constraint groups:

Constraint Groups in the IIS:

- **Minimum demand constraint** (Minimum_Premium): Requires at least 20,000 units of Premium_drink
- **Material continuity constraints:** Balance equations for cream products (Light, Medium, Heavy cream)
- **Blending constraints:** Define how Premium and Standard drinks are composed
- **Splitting/separation constraints:** Convert raw milk inputs into separated products
- **Capacity constraints:** Limit protein concentration and secondary processing throughput

Modeler Preview

User input

Incremental Specification

Questions

Test Cases

Home

Gurobot

Explainer

Modeler

API Keys

MCP

Settings

I am playing a game. We are a band of thieves and have decided to break into the famous Grandview Museum with our knapsacks and take as many items as we can carry and want to maximize the value of these items

Welcome, Steven! This sounds like a fun variation of the classic Knapsack Problem - a great fit for optimization with Gurobi.

Let me ask the most important clarifying question first:

How many thieves are in your band, and does each thief have their own knapsack with a potentially different weight capacity?

For example:

1. **Single thief** - one knapsack with a single weight capacity (classic 0/1 Knapsack)

2. **Multiple thieves, same capacity** - several knapsacks all with the same weight limit (Multiple Knapsack Problem)

3. **Multiple thieves, different capacities** - each thief has a knapsack with a different weight limit (Generalized Multiple Knapsack Problem)

Learn More:

Solving the Knapsack Problem: A Classic Challenge in Optimization | Gurobi

Knapsack Problems | Gurobi

Type your message...

Documentation

Downloads

Support

Steven Edwards

Chat files

grandview_museum_heist_specification.md (7 KB)

Grandview Museum Heist Optimization Problem

Overview

A band of thieves is breaking into the Grandview Museum. Each thief takes their own knapsack with a unique weight capacity. The museum contains a collection of items, each with a known monetary value, physical weight, and a limited number of copies available. The goal is to decide which items each thief takes in order to maximize the total value stolen, without any thief exceeding the weight limit of their knapsack. Each item copy can only be carried by one thief, and items must be taken as whole units (no splitting allowed).

Objective

Maximize the total monetary value of all items taken across all thieves' knapsacks.

Constraints

C1: Knapsack Weight Capacity

T3: Multiple Knapsacks Competing for Same Items

Description: Test scenario where multiple knapsacks could benefit from the same high-value items, but availability is limited.

Input Data:

Knapsacks:

Knapsack_ID	Capacity
Knapsack_1	10
Knapsack_2	10
Knapsack_3	10

Items:

Item_ID	Weight	Price	Max_Duplicates
HighValue	5	50.0	1
MedValue	5	30.0	5
LowValue	5	10.0	10

Acceptance Criteria:

Solution is_optimal should be True

Only one knapsack should receive the HighValue item (Max_Duplicates = 1)

Total units of each item taken ≤ Max_Duplicates

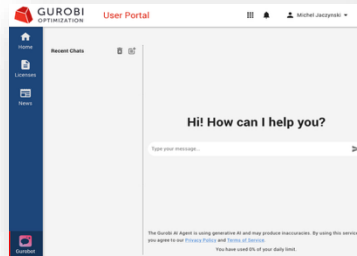
Model should efficiently distribute items to maximize total value

42

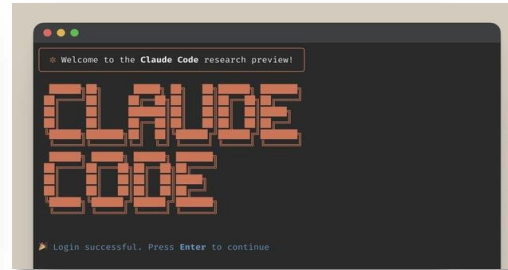
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Gurobi's Intelligence Vision

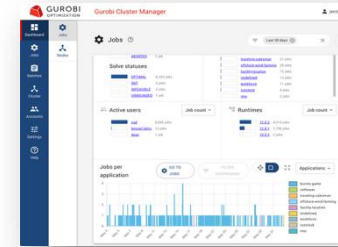
Interfaces and Integration



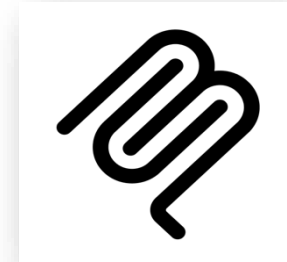
Chatbot



Coding Agents / IDEs

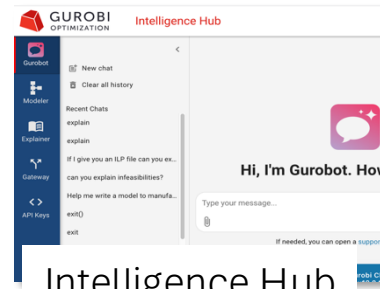


Assistants in context:
Cluster Manager, CLI

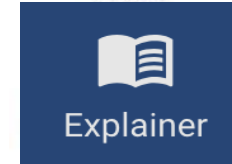
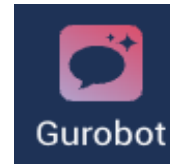


Interoperability:
MCP, A2A

Orchestration and APIs



Intelligence Hub



Agents for Gurobi related tasks:
Support, Tuning, Insight, Modeling, Solving...

Components



Knowledge Base to capture
Gurobi best practices



Gurobi Tools:
Optimizer, Tuner, Explainer
Instant Cloud, Compute Server



LLMs:
Claude, Cohere...



Thank You

For more information, visit [gurobi.com](https://www.gurobi.com)